2WikiMultiHopQA (COLING 2020)

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Abstract

This paper introduces 2WikiMultiHopQA, a large dataset for multi-hop question answering (QA). It differs from earlier datasets by requiring not only correct answers but also explanations of how each answer was reached. To build it, the authors combine Wikipedia (natural text) and Wikidata (structured triples). Each question in the dataset includes an answer, supporting sentences, and evidence triples showing the full reasoning path. The dataset is designed so that solving questions truly requires multi-hop reasoning, avoiding shortcuts that let models answer from a single paragraph.

Main Purpose

The main goal is to provide a better benchmark for evaluating multi-hop reasoning. The authors addressed three main problems in existing datasets:

- Problem 1: Many datasets only provide answers and do not explain how answers were obtained.
- Problem 2: Some "multi-hop" questions can be solved using only one paragraph, so they don't test real multi-hop reasoning.
- Problem 3: Existing datasets mostly evaluate only final-answer correctness and do not evaluate the reasoning process itself.

To fix these issues, 2WikiMultiHopQA forces genuine multi-hop reasoning, supplies complete reasoning paths as evidence triples, combines structured and unstructured data, and evaluates reasoning beyond the final answer.

Problems in Existing Datasets

The paper compares 2WikiMultiHopQA to prior datasets:

- ComplexWebQuestions: Uses a knowledge base and web pages but provides no explanations, so reasoning cannot be verified.
- QAngaroo: Built from a knowledge base, but lacks reasoning explanations and only evaluates final answers.

- HotpotQA: Provides supporting facts (sentence labels) showing where the answer came from, but these labels do not explain the reasoning steps. Researchers also showed many HotpotQA questions can be solved without multi-hop reasoning.
- R4C: Provides derivations (explanations), but it has only 4,588 questions—too small to train large models.

Because of these gaps, the authors created 2WikiMultiHopQA, which is large, includes evidence triples, ensures true multi-hop reasoning, and provides explanations.

Dataset Overview

Name: 2WikiMultiHopQA (from Wikipedia + Wikidata = 2Wiki). It merges Wikipedia articles and Wikidata triples. Each sample contains a question, context (Wikipedia paragraphs), an answer, supporting facts (sentences), and evidence triples (subject, relation, object) that form the full reasoning chain.

Example: Question: Who is the maternal grandfather of Abraham Lincoln? Context: Relevant Wikipedia paragraphs. Answer: James Hanks. Supporting facts: Sentences that contain the answer information. Evidence triples: (Abraham Lincoln, mother, Nancy Hanks) and (Nancy Hanks, father, James Hanks), which show the reasoning path from the question to the answer.

Dataset Creation Process

The dataset was generated automatically in three steps:

1. Create question templates (for example, “Who is the father of X?” or “Who died first?”).
2. Generate questions by selecting entities from Wikidata, finding related Wikipedia pages, applying templates, and producing QA pairs.
3. Post-process the generated questions by removing ambiguous ones, balancing yes/no questions, and discarding invalid reasoning chains.

Most questions are automatically generated; human workers mainly performed quality checks.

Hyperlink Graph and Bridge Entities

Unlike HotpotQA, 2WikiMultiHopQA does not build a hyperlink graph. Instead it connects Wikipedia articles to Wikidata triples to derive reasoning chains. A bridge entity is an intermediate entity linking two documents—for example, to compare directors of two movies you might move: Movie → Director (bridge) → Birth date → Comparison.

Dataset Structure and Annotation

Each sample typically contains two Wikipedia paragraphs (sometimes four). Supporting facts are the sentences that contain the needed information. Evidence is stored as triples in the form (Subject, Relation, Object), which provide a clear explanation for each reasoning step. The annotation pipeline uses templates, entity selection, Wikidata relations, and verification with Wikipedia. Human annotators mainly validated quality and filtered mismatches.

Dataset Statistics.

Category	Count
Total questions	192,606
Train	167,454
Validation	12,576
Test	12,576
Comparison	57,989
Inference	7,478
Compositional	86,979
Bridge Comparison	40,160

Question and Answer Types

There are four multi-hop question types:

- Comparison: Compare two entities (e.g., "Who was born first?").
- Inference: Apply logical rules (e.g., parent → grandparent), such as "Who is Abraham Lincoln's maternal grandfather?"
- Compositional: Solve sequential sub-questions (e.g., "Who founded the company that distributed La La Land?" requires movie → distributor → founder).
- Bridge Comparison: Combine bridge reasoning with comparison (e.g., "Which movie has the older director?" requires locating directors and comparing birth dates).

The dataset contains 708 answer types. Top types include Yes/No (31.2%), Dates (16.9%), Films (13.5%), Humans (11.7%), and Cities (4.7%). The remaining 22% cover various Wikidata entities.

Main Results and Observations

Key findings include: 2WikiMultiHopQA yields lower model scores than HotpotQA, indicating higher difficulty. Baseline performance on test F1: Answer 43.93, Supporting Facts 65.26, Evidence 14.94, Joint 5.41. Evidence generation is particularly poor, and the joint metric shows that getting everything right at once is very challenging. Human upper bounds are much higher (e.g., human answer F1 ≈ 91.79), highlighting a large gap and room for improvement.

Limitations

- Wikipedia–Wikidata mismatches: Some Wikidata facts do not match Wikipedia text, causing unanswerable or misleading questions.
- Ambiguous entity names: Entities may have multiple valid names, complicating evidence matching.
- Template-based generation: Rule-based templates can limit linguistic diversity compared to fully human-written questions.
- Remaining mismatches: Manual inspection found about 8% of sampled questions still had mismatches between Wikidata triples and Wikipedia text.